Practical Fundamentals of Geological Mapping

Introduction to terrain study in Geology

Álvaro Lozano Murciego and André Filipe Sales Mendes

Universidad de Salamanca

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CONTENTS

Welcome to the interactive book **Practical Fundamentals of Geological Mapping**.

This teaching resource has been designed for Earth Sciences students and professionals with the goal of facilitating autonomous and interactive learning of representation and terrain analysis techniques.

Book Structure

The content is organized progressively:

- ***Fundamentals of Mapping*:** Key concepts such as strike, dip, dip direction, and the interpretation of different types of geological contacts on maps.
- **Practical Exercises (Coming Soon):** A battery of solved problems and interactive Python notebooks for the automatic resolution of three-point structural problems and geological cross-sections.

PDF Version (Printable)

You can download the full version in PDF format for offline study or printing:

- *Download PDF in English*
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Note

This interactive book is a dynamic project of the **University of Salamanca (USAL)**.

Part I

Fundamentals

CHAPTER 1

INTRODUCTION TO GEOLOGICAL MAPPING

Learning Objectives

By the end of this chapter, you will be able to:

1. Understand the importance of geological mapping in Earth Sciences.
2. Identify and measure the spatial orientation of geological planes (strike, dip direction, and dip).
3. Interpret basic symbology on a geological map.
4. Solve simple conceptual problems regarding the spatial disposition of terrain.

1.1 1. What is a Geological Map?

A **geological map** is a two-dimensional, scaled representation of the spatial distribution of different rock units (or sediments) outcropping on the Earth's surface, as well as the geological structures that affect them (folds, faults, unconformities).

Unlike a conventional topographic map, a geological map provides a projection of the **third dimension** into the subsurface by interpreting the geometry of contact lines.

1.2 2. Plane Orientation: Strike, Dip, and Dip Direction

To structurally characterize any geological plane (such as a rock layer or a fault), three angular parameters measured with a geologist's compass are used:

- **Strike (Trend of the layer):** The horizontal angle between magnetic north and the line of intersection of the geological plane with a virtual horizontal plane.
- **Dip (Inclination):** The angle of maximum slope measured vertically between the horizontal plane and the geological plane. It ranges between 0° (horizontal) and 90° (vertical).
- **Dip Direction (Sense of inclination):** The direction of the horizontal projection of the line of maximum slope. It is always perpendicular to the strike (90° to the right or left).

1.2.1 Mathematical Relationship of Apparent Dip

If inclination is measured in a direction that is not perpendicular to the strike, an **apparent dip** (α) is obtained, which is always less than or equal to the **true dip** (β). The mathematical relationship is given by:

$$\tan(\alpha) = \tan(\beta) \cdot \sin(\theta)$$

Where:

- α is the apparent dip.
- β is the true dip.
- θ is the horizontal angle between the strike direction and the direction in which the apparent dip is measured.

1.3 3. Types of Geological Contacts

The geometric relationship between different rock units defines the type of contact. The most common contact relationships are outlined below:

As shown in Fig. 1.1, the diagram is versioned as a static image.

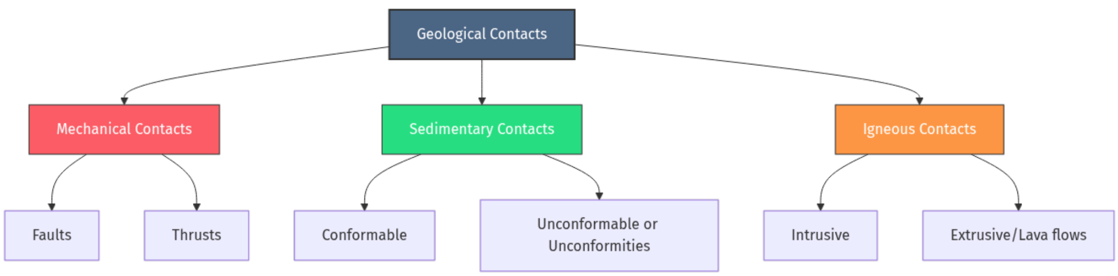

Figure1.1: Diagram generated from code.

1.4 4. The "V" Rule in Mapping

One of the fundamental rules for interpreting the spatial layout of layers from their surface exposure is the **V Rule**:

> *When an inclined layer crosses a topographic valley, the trace of its contact draws a **V** shape whose apex points in the direction of the dip of the plane (with minor exceptions where the valley slope is greater than the dip).*

- **Horizontal Beds:** Contact traces are parallel to topographic contour lines.
- **Vertical Beds:** Contact traces are represented as straight lines on the map, completely ignoring the terrain's topography.

1.5 5. Self-Assessment Exercises

Conceptual Question: Apparent Dip

Question: If a layer has a true dip of 30° towards the East, what will be its apparent dip if we observe it in a terrain cut oriented exactly in a North-South direction?

Solution: Since the dip direction is East, the strike of the layer is North-South (perpendicular to East). The direction of observation (North-South) is parallel to the strike of the layer. The angle θ between the strike direction and the direction of observation is 0°.

Applying the formula: $\tan(\alpha) = \tan(30^\circ) \cdot \sin(0^\circ) = 0 \implies \alpha = 0^\circ$

Therefore, on a cross-section plane parallel to the strike, the bed will appear completely horizontal (apparent dip of 0°).

1.6 6. Future Practical Exercises Battery

In upcoming updates to this interactive book, computational tools and executable Python notebooks will be incorporated to solve:

1. **Three-point problem:** Calculation of the strike and dip of a plane from three known elevations.
2. **Geological Cross-Section Generation:** Automated construction of cross-sections from topographic profiles and map traces.
3. **Structure contours:** Automatic tracing of contacts using spatial interpolation.

Part II

Information

CHAPTER 2

ABOUT

This book is an interactive teaching resource for **Practical Fundamentals of Geological Mapping**. It has been developed by adapting the template and materials from the course **Elaboración de libros electrónicos mediante código y asistentes de Inteligencia Artificial** at the University of Salamanca.

Citation information, BibTeX, DOI and Zenodo metadata are collected on the *How to Cite* page.

2.1 How this book is made

This website is written in Markdown files and Jupyter notebooks, and converted to HTML and PDF using tools from TeachBooks, Jupyter Book and MyST Markdown.

The source files are stored in a public GitHub repository:

- https://github.com/elloza/teachbook_usal_template

The published version of the book can be viewed at:

- https://elloza.com/teachbook_usal_template/

2.2 About the authors

Álvaro Lozano Murciego
Associate Professor, Computer Science and Automation

- USAL profile: https://produccioncientifica.usal.es/investigadores/148041/detalle

- ORCID: https://orcid.org/0000-0002-0493-4471

André Filipe Sales Mendes
Assistant Professor, Computer Science and Automation

- USAL profile: https://produccioncientifica.usal.es/investigadores/147997/detalle?lang=en
- ORCID: https://orcid.org/0000-0003-0976-2784

2.2.1 Institution

Faculties of Sciences and Chemical Sciences, University of Salamanca (USAL).

Photographs and academic profile information come from the University of Salamanca scientific production portal.

CHAPTER 3

CREDITS AND LICENSES

3.1 Project credits

This book and the course template are built with TeachBooks, on top of the Jupyter Book and MyST Markdown ecosystem.

TeachBooks provides documentation, components and working patterns that make it easier to create open, interactive and reproducible teaching books.

3.2 Acknowledgements

We thank the TeachBooks project and Delft University of Technology (TU Delft) for their work on open tools for educational books. The credits page in the TeachBooks Manual is used as a reference for explicitly acknowledging the tools, communities and institutional support on which this material builds.

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- TeachBooks
- Jupyter Book
- MyST Markdown

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CHAPTER 4

HOW TO CITE

Zenodo DOI: https://doi.org/10.5281/zenodo.20449102

If you use this material, please cite it as follows.

4.1 Recommended Citation (APA 7th edition)

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4.2 BibTeX

You can also use the downloadable BibTeX file: `citation.bib`.

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@book{lozano_murciego_elaboracion_2026,
  author = {Lozano Murciego, Álvaro and Sales Mendes, André Filipe},
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4.3 Authors

- Álvaro Lozano Murciego, ORCID: https://orcid.org/0000-0002-0493-4471
- André Filipe Sales Mendes, ORCID: https://orcid.org/0000-0003-0976-2784

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